

Training-time electrical elimination improves sparse electrothermal phase-change reconstruction beyond post-training electrical repair

Authors: [author names and affiliations to be supplied]

Corresponding author: [name and email to be supplied]

Abstract

Electrical repair at inference can restore balance without correcting a learned evolving state. We compare training-time quasi-static electrical elimination with a soft electrical PINN receiving the same final solve. A hybrid network represents temperature and phase, differentiates the electrical solution implicitly, and deposits Joule heat through the same finite-volume half-cell resistances. Explicit thermal/phase residuals and a same-solver interpolant support the comparison. In a synthetic, dimensionless two-dimensional wall cell, two initialization pairs under each of two pulse protocols favor elimination. Under the original numerical references, current/power NRMSE reductions are 48.8-78.3%, with current NRMSE 0.682-1.572% and power NRMSE 0.673-1.593%. Earlier-pulse phase RMS decreases by 20.22% and 20.31% against projected soft controls. Six additional matched phase-head continuations separate continued optimization, ordinary added capacity and a frozen interface gate; neither added head establishes the predefined increment under the original reference. Halving the reference integration step preserves 4/4 historical device advantages. One gated endpoint crosses the strict two-cycle criterion only under the refined reference, showing threshold sensitivity rather than a reference-stable capability gain. All sixteen fixed objects are independently rescored from a portable array archive. The evidence supports the specified coupling method while leaving isolated implicit-gradient causality, independent necessity of the remaining PDE terms, spatial convergence and material-calibrated performance unestablished.

Keywords: physics-informed neural network; differentiable electrical solve; electrothermal coupling; phase-field reconstruction; Joule heating; pulse history

1. Introduction

In a coupled phase-change system, a small error in potential may produce a large error in a contact current or in local heating. Conversely, repairing the electrical field can make terminal currents internally consistent while leaving an inaccurate temperature or phase distribution unchanged. These distinctions matter when a neural reconstruction is assessed through device quantities rather than field images alone. Our question is whether the electrical constraint should participate in learning the evolving state, or whether a final electrical solve is sufficient.

We examine this question in an observation-driven reconstruction task. Sparse potential, temperature and phase samples, the geometry, the drive and all constitutive parameters are known. This is deliberately not a claim that a neural model is necessary to solve a fully specified forward problem: a conventional coupled solver can do so without interior observations. The task instead provides a controlled setting for comparing how alternative

physics interfaces assimilate the same limited field information. The reference is a fixed numerical trajectory, not an experiment or a continuum solution.

Physics-informed neural networks combine data with differential-equation residuals [1]. Such losses do not automatically ensure that every field, constraint and functional quantity improves together. Gradient imbalance and network parameterization can substantially affect optimization [2]. Differentiable numerical components offer another way to impose selected constraints. Solver-in-the-Loop trains neural corrections through a differentiable simulator and emphasizes how learning interacts with its numerical trajectory [3]. Hybrid finite-element/neural models instead embed neural constitutive components in a PDE-constrained formulation [4]. Modular implicit differentiation supplies a general means of differentiating a solution defined by an optimality or root condition [5]. These precedents motivate the tools used here; neither implicit differentiation nor inserting a solver into learning is claimed as a new principle.

Reduced and all-at-once PDE formulations distinguish eliminating a constrained variable from optimizing the full state jointly [13]. Integral projection methods enforce prescribed global quantities [12]; our electrical layer instead solves a local quasi-static boundary problem, retaining the evolving thermal and phase fields as neural unknowns. These neighboring approaches delimit the contribution, rather than establish novelty for constrained learning itself.

Our setting differs in three respects. First, only the quasi-static electrical variable is eliminated; the transient temperature and phase fields remain neural functions constrained by observations and explicit residuals. Second, the electrical operator and local Joule deposition share the same face resistances, so the heat source used during learning is consistent with the electrical discretization. Third, the comparison gives the soft electrical model the same final electrical solve. This last control separates a benefit from learning with the constraint from a benefit explained entirely by correcting potential at evaluation. A full-spatial-penalty counterfactual and two clean initialization pairs further delimit the interpretation.

Control-volume PINNs have previously constructed residuals from integrated conservation laws, including additional regularization for hyperbolic systems [6]. We use a spatial cell-balance thermal residual with automatic differentiation at shared faces; we do not inherit a space-time entropy-stability guarantee. The phase dynamics are of nonconserved phase-field type [7]. The wall-cell layout and electric-thermal-phase feedback are inspired by multiphysics phase-change-memory modeling [8], but the present coefficients are dimensionless design values. In particular, that application source concerns Ge-rich GST, not an oxide calibration. Our numerical model omits its multi-phase composition, material interfaces and field-dependent switching physics.

The contribution is therefore a defined coupling method and a controlled evidence sequence. We first distinguish fixed-state electrical repair from remaining differences after a common repair. We then examine whether the benefit persists under fresh initialization and a finite change in pulse history. Finally, we report the event and energy counterexamples that constrain the method's usefulness. The aim is a testable computational-method claim, rather than a collection of individually named optimization techniques.

2. Two-dimensional model and reconstruction task

2.1 Governing equations

The computational cell is $\Omega = [-1,1] \times [0,1]$, with coordinates x and z and time $0 \leq t \leq 2.5$. Potential V , reduced temperature T and phase fraction ϕ obey

$$\nabla \cdot (\sigma(T, \phi) \nabla V) = 0. \quad (1)$$

$$T_t + L\phi_t = \alpha \Delta T - \gamma T + Q\sigma(T, \phi)|\nabla V|^2. \quad (2)$$

$$\phi_t = M(T)[\varepsilon^2 \Delta \phi - \partial_\phi W(\phi, T)]. \quad (3)$$

The conductivity, mobility and local phase potential are

$$\sigma = \exp[\beta T + \log(R)\phi^2(3 - 2\phi)], \quad (4)$$

$$M(T) = m_c + (m_h - m_c) \text{sigmoid}[(T - T_c)/w_M], \quad (5)$$

$$W = B\phi^2(1 - \phi)^2 + D(T_c - T)\phi^2(3 - 2\phi). \quad (6)$$

We use $\alpha = 0.10$, $\gamma = 4$, $L = 0.05$, $Q = 4$, $\beta = 0.25$, $R = 8$, $\varepsilon = 0.04$, $B = 1$, $D = 6$, $T_c = 0.45$, $m_c = 0.5$, $m_h = 5$ and $w_M = 0.08$. Mobility multiplies the entire phase bracket; it is not moved inside a divergence. Because W depends on the externally evolving temperature, the model is not assigned an autonomous monotone free-energy constraint.

The full top electrode is driven by $U(t)$; a centered bottom segment $|x| \leq 0.35$ is grounded. All remaining electrical faces are insulating. The top temperature is zero; the other thermal boundaries satisfy $\partial_n T + 0.25T = 0$. Phase satisfies $\partial_n \phi = 0$ on every boundary. Initially $T = 0$ and

$$\phi_0 = 0.02 + 0.01 \exp \left[-\frac{1}{2}((x/0.18)^2 + ((z - 0.12)/0.10)^2) \right]. \quad (7)$$

These assumptions define a synthetic wall-cell benchmark. No dimensional material parameters, experimental memory operation, or oxide-specific constitutive identification are inferred from it.

2.2 Complete pulse protocols and information boundary

Each pulse rises linearly from zero to 0.72 over its first 0.05 time units, holds until local time 0.27, then falls to zero at 0.35. The original protocol starts two pulses at 0 and 1.25. The intervention changes only the second start to 1.01, leaving the pulse shape, total time, coefficients and boundaries fixed. The input contains exactly two pulses; it is not periodically extended to create a third pulse. Advancing the second pulse by 0.24 preserves its phase relative to the 0.02 observation spacing.

Support trajectories use 80×40 cells, a main time step of 0.0025 and saving every two steps. The spatial and temporal masks take indices 0,4,8,... and the last index, fixed independently of field values. The resulting $21 \times 11 \times 126$ mask contains 231 analytic initial positions and 28,875 positive-time positions, with 86,625 scalar field labels. Training receives only these samples, coordinates and known physical laws. Dense support fields, reference fields, reference currents or heating, and sealed stress cases do not enter training or model selection.

The evaluation reference uses 160×80 cells, a main time step of 0.000625 and saving every four steps, giving 1001 times. These discretizations serve different roles; their existence alone is not a convergence study. Each protocol has its own observations and is fitted separately. The intervention is consequently a complete-case adaptation experiment, not zero-shot transfer or a formal out-of-distribution test. The two seed values are reused across protocols for paired reporting, not counted as four independent physical cases.

3. Partially eliminated hybrid PINN

3.1 Neural states and electrical elimination

Let $u_\theta = (T_\theta, \phi_\theta)$. Independent smooth modified-MLP heads [2] have width 64 and four hidden layers; the shared temperature architecture also includes a small additive latent adapter described in the Supplement. FP64 arithmetic and identical output mappings are used in both methods. With $a(t) = 1 - \exp(-t/0.35)$,

$$T_\theta = 2.5a(t)(1 - z)\text{sigmoid}(h_T), \quad (8)$$

$$\phi_\theta = \text{sigmoid}[\text{logit}(\phi_0) + 8a(t)h_\phi]. \quad (9)$$

The eliminated method, E, evaluates σ_θ from these states and solves the grounded electrical system

$$A(\sigma_\theta)v^* = f(U, \sigma_\theta). \quad (10)$$

For an internal face e between cells i and j , the half-cell distances d and face area A_e define

$$R_{ei} = d_{ei}/(\sigma_i A_e), \quad R_{ej} = d_{ej}/(\sigma_j A_e), \quad g_e = (R_{ei} + R_{ej})^{-1}. \quad (11)$$

Electrode faces use the corresponding half-cell resistance and the actual heater overlap. Insulating faces contribute no outward conductance. Positive conductivity and the grounded electrode yield a nonsingular discrete electrical problem. The implementation uses sparse factorization, not a dense inverse. For a scalar loss, the complete first-order derivative is obtained from

$$A^T z = \partial \mathcal{L} / \partial v, \quad d\mathcal{L} = d\mathcal{L}_{\text{direct}} + z^T(df - dA v^*). \quad (12)$$

The direct term includes the explicit dependence of heating on conductivity. The right-hand side f also depends on electrode conductance, and its derivative is retained. A factorization may be reused for the adjoint of the same forward solve, but not after changing the state. This is an application of implicit differentiation [5], with no claim of a new differentiation theorem or support for arbitrary second-order derivatives.

3.2 Consistent local Joule deposition and remaining residuals

Given either the solved or neural potential, face current and deposition are

$$I_e = g_e(v_i - v_j), \quad P_{e \rightarrow i} = I_e^2 R_{ei}, \quad P_{e \rightarrow j} = I_e^2 R_{ej}. \quad (13)$$

Electrode power enters the neighboring boundary cell. For cell volume ω_i , q_i is its deposited power divided by ω_i . This allocation is not generally an equal split between adjacent cells. It gives

$$\sum_i \omega_i q_i = P_j, \quad (14)$$

where P_j sums all internal and electrode dissipation. Equation (14) is an accounting identity for the shared face network, even when a neural potential violates electrical balance. Equality with terminal power UI additionally requires the electrical equations. Neither identity by itself establishes accuracy relative to the reference heating field.

The temperature residual uses a cell balance,

$$R_{T,i} = \partial_t(\bar{T}_i + L\bar{\phi}_i) + \gamma\bar{T}_i - \frac{\alpha}{\omega_i} \sum_f A_f (\nabla T_\theta)_f \cdot n_{if} - Qq_i. \quad (15)$$

Cell means use midpoint quadrature. Temperature gradients are differentiated at shared face nodes with opposite normals for neighboring cells, including the model's own gradients on boundary faces. The same thermal boundary residual remains in both losses. The factor Q is applied once. The phase residual is

$$R_\phi = \phi_t - M(T)[\varepsilon^2 \Delta \phi - 2B\phi(1-\phi)(1-2\phi) - 6D(T_c - T)\phi(1-\phi)]. \quad (16)$$

Thus the thermal interface is a spatial cell-balance residual and the phase interface is a pointwise AD residual. They are not claimed to equal the previous pointwise thermal formulation or the full space-time construction of [6]. At $U = 0$, E uses the analytic electrical solution $v = q = 0$ while thermal and phase evolution remain active.

3.3 Soft comparator and training objectives

The soft electrical PINN, F, trains all three field heads. Its potential is $V_\theta = U(t) \exp(h_v)/[\exp(h_v)+1-z]$, preserving the voltage range and the exact top value. It evaluates the same face operator and Joule deposition as E, but penalizes

$$r_{e,i} = (A(\sigma_\theta)V_\theta - f(U, \sigma_\theta))_i/\omega_i. \quad (17)$$

No additional, differently averaged electric boundary loss is added: electrical boundary conductances already enter this residual. Let L_{obs} denote the common weighted observation loss, including the full initial-logit increment for phase, and let L_{BC} and L_{IC} be the common remaining boundary and initial losses. Their exact measures are specified in Supplement S2. With one calibration a, b at the shared parent and λ rising to 0.1,

$$J_{T\phi} = \frac{1}{3} \mathbb{E}_\rho [(R_T/4)^2 + (R_\phi/5)^2], \quad (18)$$

$$\mathcal{L}_E = L_{\text{obs}, E}/a + \lambda(5L_{\text{BC}} + L_{\text{IC}} + J_{T\phi, E})/b, \quad (19)$$

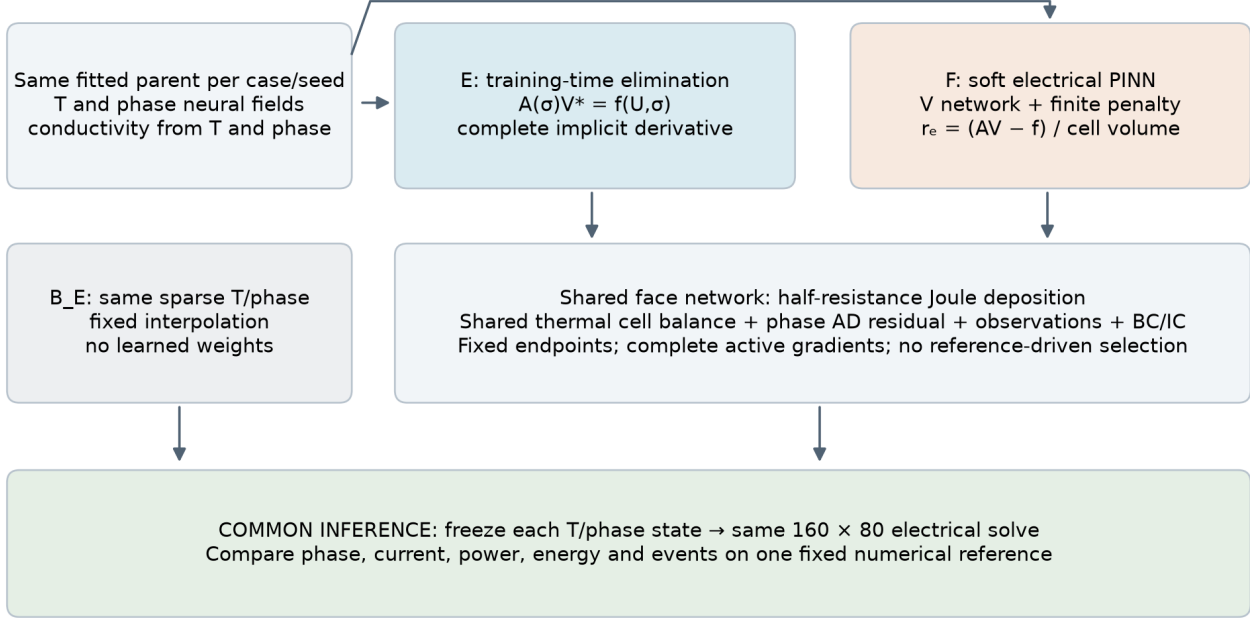
$$\mathcal{L}_F = L_{\text{obs}, F}/a + \lambda(5L_{\text{BC}} + L_{\text{IC}} + J_{T\phi, F})/b + \frac{\lambda\eta}{3b} \mathbb{E}_\rho [r_e^2]. \quad (20)$$

Here ρ is the uniform space-time target with the declared time-window quadrature. Scales a and b are shared within a pair and frozen, with a numerical floor of 10^{-12} . The primary F uses $\eta = 1$. A development control uses a one-time gradient-balanced η ; a second control replaces the 128 sampled electrical cells by the volume mean over all 3200 cells at each physical time. The latter changes spatial integration, not the penalty by a factor of 25. Thermal and phase terms keep their original interfaces.

E freezes the unused potential head. Its voltage observations differentiate through conductivity and the solve, whereas F voltage observations act on its potential head. E

enforces its constraint at powered observation and physics times; F penalizes it at physical residual times. Initial potential, active parameters, constraint strength and gradient maps consequently differ. These are explicit parts of the method-package comparison, not hidden evidence for an isolated VJP mechanism.

Sparse observations + known geometry, drive, constitutive laws and IC/BC



Method-package comparison: initial V, active parameters, constraint times and gradient maps differ.
The design does not isolate the implicit derivative alone. T/phase and their events are unchanged by final projection.

Figure 1. Both methods use the same sparse observations and evolving-state architecture. Common inference freezes each learned T/phase state and solves the same electrical problem. The interpolant B_E also receives this solve. The schematic describes implemented interfaces, not a new experiment; implementation sources are mapped in Supplement S7.

3.4 Matched development of the phase representation

The main elimination study leaves incomplete first-cycle support recall despite accurate electrical readouts. We test the hypothesis that a limited phase parameterization contributes to this mismatch. The hypothesis does not establish sigmoid saturation as its cause, nor imply that adding parameters must help. The two final earlier-pulse E states are used as separate parents. This is subsequent development on two already trained states, not another pair of clean initialization confirmations.

An unchanged continuation, E_C , is compared with an ordinary residual head, E_R , and a parent-interface-modulated residual head, E_I . These identifiers denote methods; the current error in the evaluation equation is a separate functional quantity. Let the original complete logit be $\square_\theta = \text{logit}(\phi_0) + 8a(t)h_\phi \cdot \theta$. The residual R_ξ takes the original normalized coordinates as input and uses a 3-32-32-1 tanh MLP with 1217 trainable parameters. The output weight and bias are initially zero. E_R and E_I use identical initial residual tensors, with seeds 916029 and 916043 for the two parents. Their complete initial fields exactly equal those of E_C .

For E_I , a frozen copy of the parent's complete phase function defines

$$g = 0.25 + 0.75[4\phi_{\text{parent}}(1 - \phi_{\text{parent}})], \quad c_g = \sqrt{\mathbb{E}_\rho[g^2]}, \quad \hat{g} = g/c_g. \quad (21)$$

The positive floor leaves a correction path outside the parent's predicted interface, including possible missed regions. The scalar c_g is calculated once using the parent's original unlabeled calibration quadrature; it equals 0.2693107755 and 0.2705578194 for seeds 29 and 43. It controls an RMS amplitude, not every gradient norm or the effective optimizer step. The outputs are

$$\phi_R = \text{sigmoid}(\ell_\theta + 8aR_\xi), \quad \phi_I = \text{sigmoid}(\ell_\theta + 8a\hat{g}R_\xi). \quad (22)$$

Only the gate's parameters are frozen. Its coordinate derivatives, including second spatial derivatives, remain in the phase residual. The two added residuals have equal trainable counts, but E_I has extra frozen parameters and extra function/coordinate-derivative evaluations; it is not equal to E_R in total model storage or work. No logit is recovered by inverting an already saturated phase prediction. One composed phase interface is used by the observation increment, conductivity, electrical solve, Joule deposition, thermal and phase equations, boundaries, initial conditions and inference. Thus the comparison changes representation while preserving the actual coupled PINN objective.

Residual parameterizations have established precedents [11], and gated mixing is already present in the modified-MLP background [2]. The present residual head is neither the original ResNet architecture nor a claim to invent gating. E_C -to- E_R assesses added capacity and its optimization effects. E_R -to- E_I assesses the particular frozen-gate parameterization, including its coordinate-derivative cost. A gain over E_C alone cannot establish an independent gate benefit.

3.5 Fixed-prediction reference perturbation

Both protocols receive an additional reference with the original algorithms and tolerances, the same 160 by 80 spatial grid, and half the integration step: 0.0003125 with saving every eight steps. Original references remain available. The 1001 saved times, predictions, ROI and event definitions are identical between the two scoring passes; no prediction is interpolated or reselected. New references are excluded from training and method selection. Each normalized error uses its own reference denominator, with a fixed-old-denominator diagnostic reported separately.

For an error in one common weighted norm, the reverse triangle inequality gives $|e_{\text{new}} - e_{\text{old}}| \leq \delta$, where δ is the norm of the reference difference. Consequently, the relative-gain margin obeys

$$m = 0.9e_F - e_E, \quad |m_{\text{new}} - m_{\text{old}}| \leq 1.9\delta. \quad (23)$$

The same bound applies to active-set symmetric difference using δ_S , the measure of the two reference sets' symmetric difference. Normalized device margins use a fixed denominator for this bound; a changed denominator is reported separately. These bounds contextualize a narrow margin and do not prove that every threshold decision is robust. Temporal refinement alone does not establish spatial convergence or continuum accuracy.

4. Matched experiment and evaluation

4.1 Shared parents, fixed endpoints and strong controls

For each protocol, seeds 29 and 43 initialize fresh networks and a zero-output adapter, without loading trained weights, optimizer history or calibration. A common observation-only parent receives 2400 Adam updates [9] and 600 full-observation L-BFGS evaluations, 200 per field head [10]. E and F then start from that same fitted parent with fresh optimizers. Each branch receives 1500 Adam updates and at most 300 fixed complete-objective/gradient evaluations. Every line-search trial and repeated evaluation counts; budget termination restores the last accepted state. No reference score selects a checkpoint.

The learning rate, Adam parameters, clipping, observation stream, physical time windows and residual scales are shared. L-BFGS uses fixed data and quadrature, without resampling or gradient clipping inside its closure. Equal update and evaluation caps do not imply equal computation: E incurs sparse forward and adjoint solves, while F differentiates a potential network and its explicit residual. Actual work is retained in Supplement S6; no speedup is claimed.

B_E interpolates the same sparse temperature and full-logit phase information using fixed PCHIP and spatial interpolation rules, then solves the same electrical system. It is computed once per protocol and reused across seeds. B_E has access to the same three-field sparse carrier, but its state interpolation actually uses only T and phase; it does not assimilate interior V observations. This is a comparison with the same available data, not equal utilization of every observed field. Historical development controls include D_E , which omits the remaining thermal/phase interior residuals, and the balanced and full-spatial versions of F. They are reported separately, not pooled with clean repetitions. The primary soft recipe was locked before the clean confirmation rather than selected anew for each seed.

4.2 Common inference and distinct outcome measures

All final projected states use the same 160×80 electrical layer at the reference times. The unprojected F readout is retained separately. For a fixed F, projection changes V, q and electrical readouts while leaving T, ϕ and every phase event unchanged. The error measures distinguish these effects.

S is the space-time mean symmetric difference of active sets, where active means $\phi \geq 0.5$; it uses the full cell domain and global trapezoidal time weights. Raw E_ϕ is phase RMS on the ROI $|x| \leq 0.55$, $0 \leq z \leq 0.55$. E_T is ROI temperature RMS divided by 0.45, and E_V is unnormalized full-domain potential RMS. Device errors are

$$\mathcal{E}_I = \frac{\|I - I_{\text{ref}}\|_t}{\|I_{\text{ref}}\|_t}, \quad \mathcal{E}_P = \frac{\|P_J - P_{J,\text{ref}}\|_t}{\|P_{J,\text{ref}}\|_t}, \quad \mathcal{E}_W = \frac{|\int (P_J - P_{J,\text{ref}}) dt|}{|\int P_{J,\text{ref}} dt|}. \quad (24)$$

The time norm is trapezoidal RMS over the complete interval. The matched device criterion uses bottom current; top and bottom currents agree to solve accuracy for projected states. Percentages below are 100 times normalized errors, not dimensional percentage errors in temperature or phase. All absolute, percentage-point and relative differences are provided in the unified tables.

For the same heating-window measure, let M_{pred} , M_{ref} and M_{TP} be predicted, reference and overlapping active masses. When both masses exceed the numerical denominator floor, the definitions give

$$\text{recall} = \text{precision} \frac{M_{\text{pred}}}{M_{\text{ref}}} \leq \min \left(1, \frac{M_{\text{pred}}}{M_{\text{ref}}} \right). \quad (25)$$

This identity separates an insufficient predicted active mass from mismatch in space and time at a fixed mass. It is a descriptive support diagnostic, not a new loss or evidence that a particular neural mechanism caused the error. The original near-zero floors remain in the scorer. The strict gates are project-defined evaluation requirements, not experimental device-reliability standards or estimates of a success probability.

Predeclared phase-reconstruction and device-function criteria require 10% improvements in their named primary metrics with 5% noninferiority guards and fixed absolute floors. They are retained in Supplement S3, rather than used as a substitute for reporting error magnitudes. Failure to cross an advantage threshold does not demonstrate equivalence. Strict event usability is separate: both cycles must meet onset error ≤ 0.005 , recall ≥ 0.9 , precision ≥ 0.8 , mass ratio 0.8-1.2, and the specified peak, locality and recovery conditions. An onset is the linearly interpolated upward crossing of ROI active fraction 0.02. Support metrics integrate cell-volume overlap during each heating window.

For projected states, $P_J = \text{UI}$ to electrical solve accuracy. Current and power errors thus differ in drive-dependent temporal weighting, but are not independent physical replications. Likewise, a small integrated energy error may reflect cancellation. We therefore retain the signed per-pulse errors, event metrics and local state errors even when global device criteria pass.

4.3 Phase-head continuation and independent array scoring

For each earlier-pulse seed, all three continuation arms open the existing T/phase parameters and temperature adapter; E_R and E_I additionally open their residual parameters. The unused V head remains frozen. The original seed's a,b, observation measures, thermal/phase scales 4/5 and denominators 3/13 are unchanged. Lambda is fixed at 0.1 from the first update. Each arm receives 600 fresh-Adam updates at learning rate 0.0001 followed by at most 100 complete fixed-target L-BFGS evaluations. The six-arm ceiling is 3600 updates and 600 evaluations. All three arms share the same seed-specific observation and physical sampling streams. No arm is rescued using reference results.

The endpoint is the fixed-budget final accepted state or the optimizer's intrinsic stopping state. All line-search trials are counted and rejected trial states are rolled back. Electrical solve limits are 11400 forward and 11400 adjoint solves per arm. Complete evaluation uses all 126 observation times and the original 32 physical times; at most 34 observation times have nonzero drive. The six new projected predictions require at most 1668 electrical solves. Actual work, extra gate evaluation and coordinate differentiation are reported in Supplement S10.

A representation increment requires the original phase or device criterion against E_C and the corresponding noninferiority to the original E: temperature, top-current and potential errors for the phase criterion; S, phase, temperature, potential and top-current errors for the device criterion. Additional checks of all errors against the parent are reported separately, not added as new eligibility gates. Separately, a first complete strict two-cycle success may be reported when S, phase, temperature, potential, current and power errors remain within the original noninferiority limits against both controls; this does not

retroactively pass a failed A/B criterion. Gate independence additionally requires the same kind of gain against E_R , which has the same trainable parameter count. Mixed seed outcomes remain a development signal, not a stable confirmation. The old projected soft models and B_E remain in the tables.

The old soft models do not receive this extra continuation budget. Their rows remain useful reference points, but a new head versus an old F is not an equal-budget network ablation. The new representation attribution comes from $E_C/E_R/E_I$, whereas the original clean E/F experiment retains its own matched budget and identity.

The independent scoring package contains all fixed arrays and serialized metric definitions. Its evaluator uses only NumPy: it does not load a neural checkpoint or execute a linear solver. It first reproduces the ten historical objects' saved metrics, events and decisions, then evaluates the six new objects and the refined references. Figure rebuilding, array-only rescoring and retraining are separate operations. No local archive is represented as a public dataset or assigned an invented identifier.

5. Results

Sections 5.1-5.5 retain the historical results against the original fixed references. Section 5.6 adds the matched phase-head development, and Section 5.7 separately reports the new reference perturbation. No historical endpoint or decision is overwritten.

5.1 Electrical repair is strong but does not explain the full method difference

The development comparison first demonstrates how misleading uncorrected electrical outputs can be (Figure 2). Holding the soft model's T and ϕ fixed, the final solve reduces F_{raw} current NRMSE from 279.30% to 1.071% and power NRMSE from 20.118% to 1.065%. The full-spatial soft model also improves substantially on projection, from 38.754% to 1.187% current error and from 6.611% to 1.190% power error. None of these changes is a phase-learning improvement: the phase field is unchanged.

After every state receives the common solve, E still has 0.694% current and 0.682% power error, lower than both soft models. Against projected F_{raw} , the absolute reductions are 0.377 and 0.383 percentage points, respectively. This remaining difference cannot be attributed entirely to withholding a final solver from the comparator. The values are nevertheless sub-percentage-point changes in a dimensionless numerical benchmark, not measured energy savings or yield gains in an experimental device.

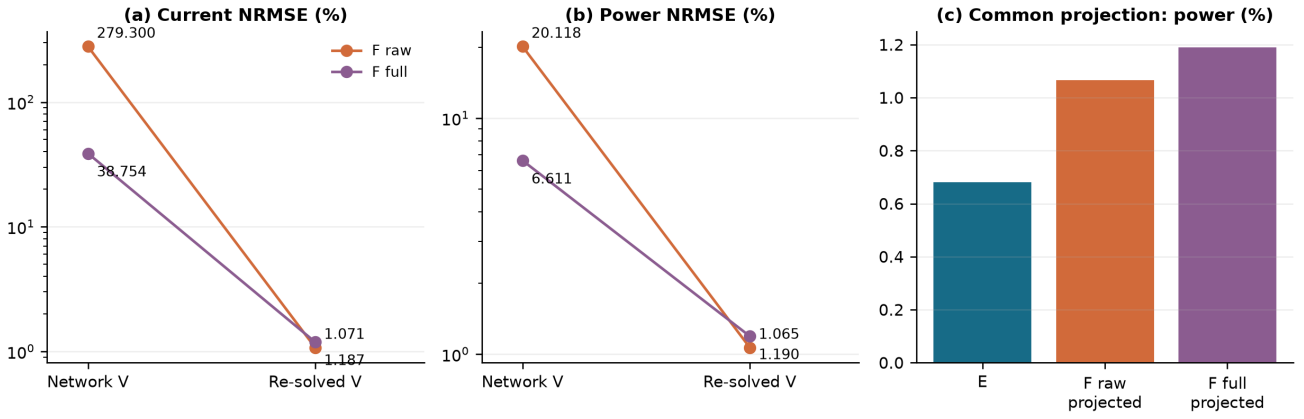


Figure 2. Panels a-b pair each fixed soft state before and after electrical re-solving; logarithmic vertical axes expose the large repair. Panel c compares projected endpoints, including E. T/phase and event metrics are unchanged within each repair pair. Source: the saved development-control table consolidated in Supplement S4; these states are not additional clean initialization repetitions.

5.2 Full spatial coverage does not remove the development signal

The full-grid electrical mean retains the same global scaling as the sampled mean. It tests whether sparse spatial enforcement alone explains the observed difference. In this development comparison, projected F_{full} remains worse than E in current and power, while its integrated energy error is better: 0.0362% versus 0.2188% (Figure 3). A single integrated quantity would therefore give a different ranking. The balanced penalty also does not outperform E on the functional metrics.

D_E is an important counterexample to a stronger interpretation. It obtains lower phase RMS than E, 0.01558 versus 0.01600, and nearby current/power errors of 0.724%/0.721%. A subsequent same-parent residual-strength study did not establish the frozen independent increment from the remaining thermal/phase PDE terms. The current hybrid PINN contains those residuals, but their necessity is not a supported contribution. The full-spatial experiment also leaves differences in enforcement times, finite versus exact constraints and initial V; it does not establish superiority to every sufficiently optimized soft formulation.

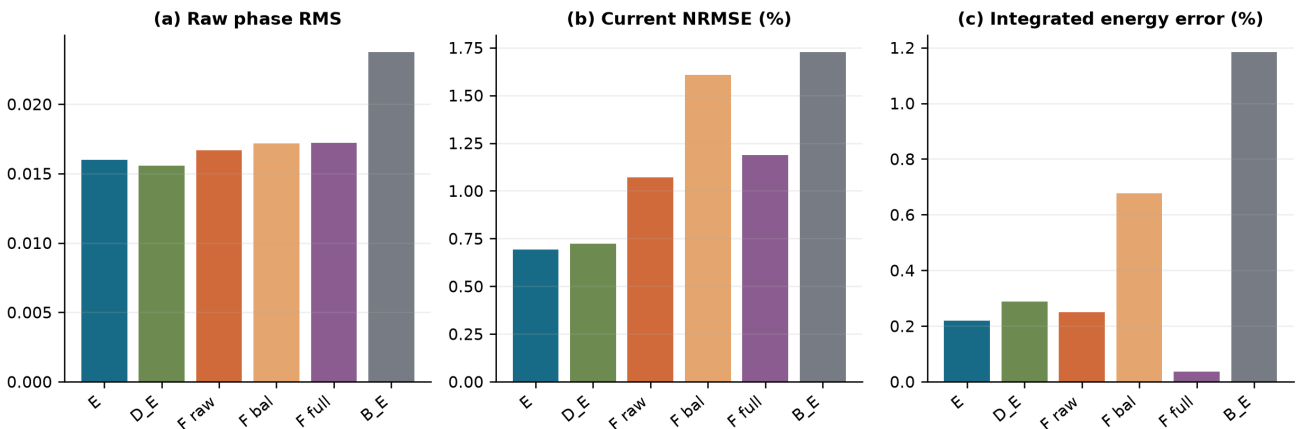


Figure 3. All shown device outputs receive the common solve. D_E omits thermal/phase interior residuals; F_{raw} , F_{bal} and F_{full} retain them and vary the electric penalty. B_E is the same-solver interpolant. F_{full} integrates the electrical residual over all cells with a volume mean. Its favorable energy error and the strong D_E phase result delimit the claim. Sources and full metrics are in Supplement S4.

5.3 Clean pairs preserve device benefits under both protocols

Tables 1-2 and Figure 4 report the two protocols and two initialization values individually. B_E appears once per case. Under the original protocol, E reduces current error relative to projected F by 49.15% and 48.80%, and power error by 50.07% and 49.85%. Under the earlier second pulse, these reductions are 56.20%/77.20% for current and 57.21%/78.26% for power. The corresponding absolute new-protocol reductions are 1.276/2.310 percentage points in current and 1.336/2.422 points in power.

Table 1. State reconstruction at the fixed endpoints. S and potential RMS are displayed after multiplication by 1000; phase RMS is raw. Temperature error is normalized by 0.45 and expressed as a percentage. All values are relative to the fixed numerical reference, not continuum truth.

Case	Seed	Method	S ($\times 10^{-3}$)	Phase RMS	T RMS / 0.45 (%)	V RMS ($\times 10^{-3}$)
Original	29	E	1.03008	0.0200421	1.06007	1.29581
Original	29	F/projected	1.225	0.0239867	1.09308	2.60377
Original	43	E	1.13164	0.0216192	1.03308	2.35729
Original	43	F/projected	1.2525	0.0244821	1.04467	4.64753
Original	--	B_E	1.40844	0.0237698	1.32783	2.55845
Earlier pulse	29	E	1.08617	0.0190024	1.10217	1.5681
Earlier pulse	29	F/projected	1.26125	0.0238185	1.16914	3.72351
Earlier pulse	43	E	1.07352	0.0186192	1.0961	1.08025
Earlier pulse	43	F/projected	1.19773	0.0233632	1.14088	4.58526
Earlier pulse	--	B_E	1.57547	0.0255235	1.28002	3.00566

Table 2. Device reconstruction after a common electrical solve. Errors are percentages. Each interpolant is one shared deterministic baseline, not a separate seed-level experiment.

Case	Seed	Method	Current NRMSE (%)	Power NRMSE (%)	Energy error (%)
Original	29	E	0.858273	0.862489	0.275172
Original	29	F/projected	1.68784	1.72743	0.160928
Original	43	E	1.57178	1.59275	0.656891
Original	43	F/projected	3.06966	3.17575	1.21912
Original	--	B_E	1.72823	1.75623	1.18494
Earlier pulse	29	E	0.994297	0.999332	0.287606
Earlier pulse	29	F/projected	2.26991	2.33532	0.706627
Earlier pulse	43	E	0.682119	0.672758	0.162718
Earlier pulse	43	F/projected	2.99173	3.09469	0.25736
Earlier pulse	--	B_E	2.00319	2.04894	1.50593

All four E/soft pairs pass the prescribed device-function criterion. Both earlier-pulse pairs also pass the phase-reconstruction and device criteria against B_E . On the original protocol, seed 43 does not cross the strong-interpolant advantage gate: its phase, current and power reductions are only about 9.05%, 9.05% and 9.31%. This failure remains relevant despite

favorable directions. The historical seed 17 used a different development history and is not pooled into these confirmations.

Raw phase reductions against projected F are 16.45% and 11.69% for the original protocol, and 20.22% and 20.31% for the earlier pulse. The earlier-pulse seed-43 S reduction is 10.3711%, leaving an absolute margin of only 4.4453×10^{-6} beyond the 10% criterion. That individual threshold decision has a narrow margin under this reference; it is not evidence of robustness to reference discretization. No significance test is based on the number of grid cells or times, and two initializations do not establish a population success probability.

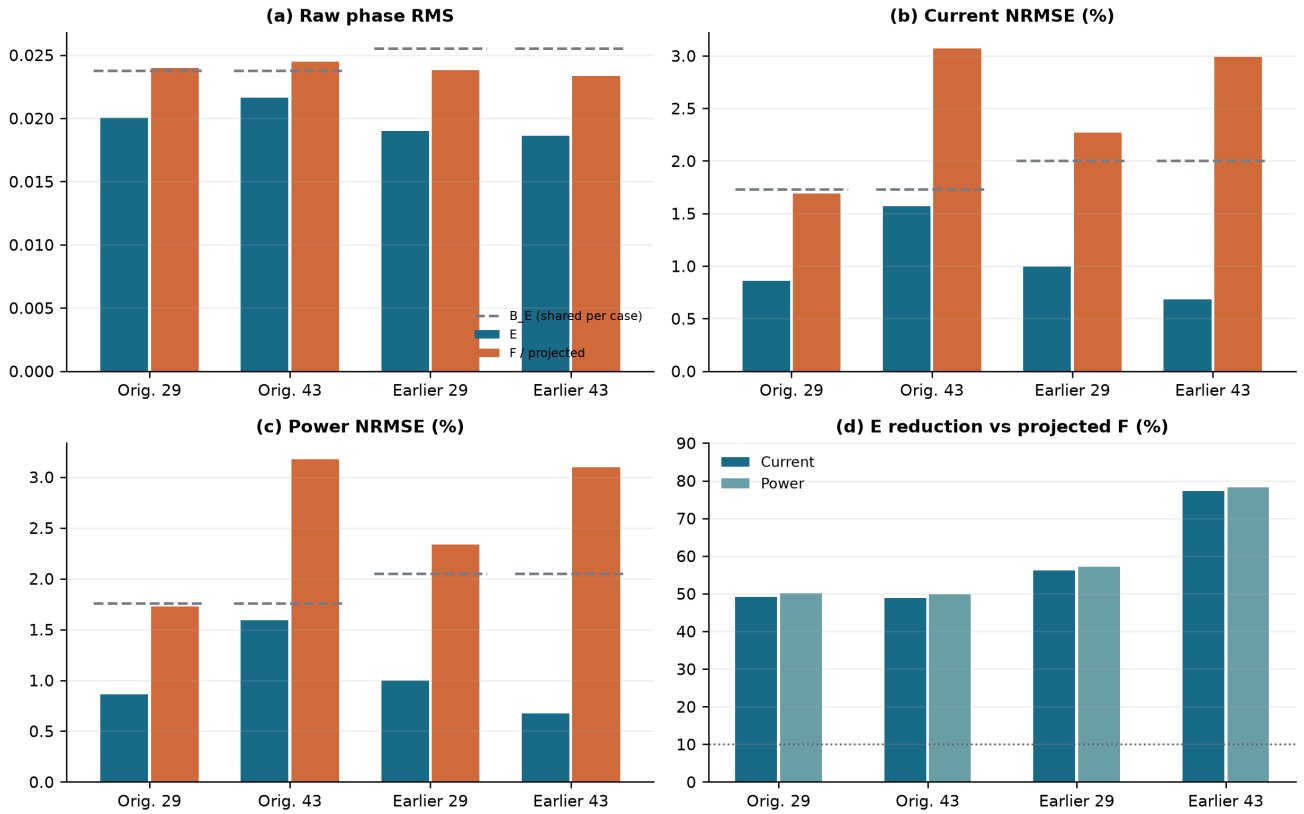


Figure 4. Each group is a separately fitted E/F pair; the two cases reuse initialization identifiers. Dashed B_E values are repeated visually for comparison but counted once per case. The fourth panel gives relative reductions, while panels a-c retain absolute error scales. Sources: unified results and paired-effects tables; no newly selected checkpoint or rerun is included.

5.4 Threshold recovery does not imply a reset of the continuous state

The fixed numerical references agree before the intervention to a maximum reported field difference of 7.22×10^{-15} at matching resolution. Immediately before the second pulse, however, the earlier protocol has a warmer and less relaxed state: ROI mean T changes from 0.006525 to 0.018746, and mean ϕ from 0.0004829 to 0.0039725. Maximum pre-pulse ϕ increases from 0.02056 to 0.19179, still below the active threshold. The threshold-based recovery fraction is therefore compatible with appreciable continuous-state memory.

The reference second-event latency decreases from 0.2484 to 0.2168, a shortening of 0.0316, or 12.72% of the original latency. A report-only calculation from the saved event table gives predicted shortenings 0.027942 and 0.030740 for E29/E43, 0.035000 and 0.015580 for F29/F43, and 0.025271 for B_E . The corresponding absolute shortening errors

are 0.003658, 0.000860, 0.003400, 0.016020 and 0.006329. F29 is slightly better than E29 on this difference, even though its device trajectories are worse.

Figure 5 connects residual states, timing changes and electrical response. The intervention supports a pulse-history effect within this model; it does not isolate thermal from phase-memory mediation. Each neural function is fitted offline using observations over its complete trajectory. Different neural predictions over the physically identical prefix therefore reflect separate reconstruction errors, not a violation of the generator's causal evolution.

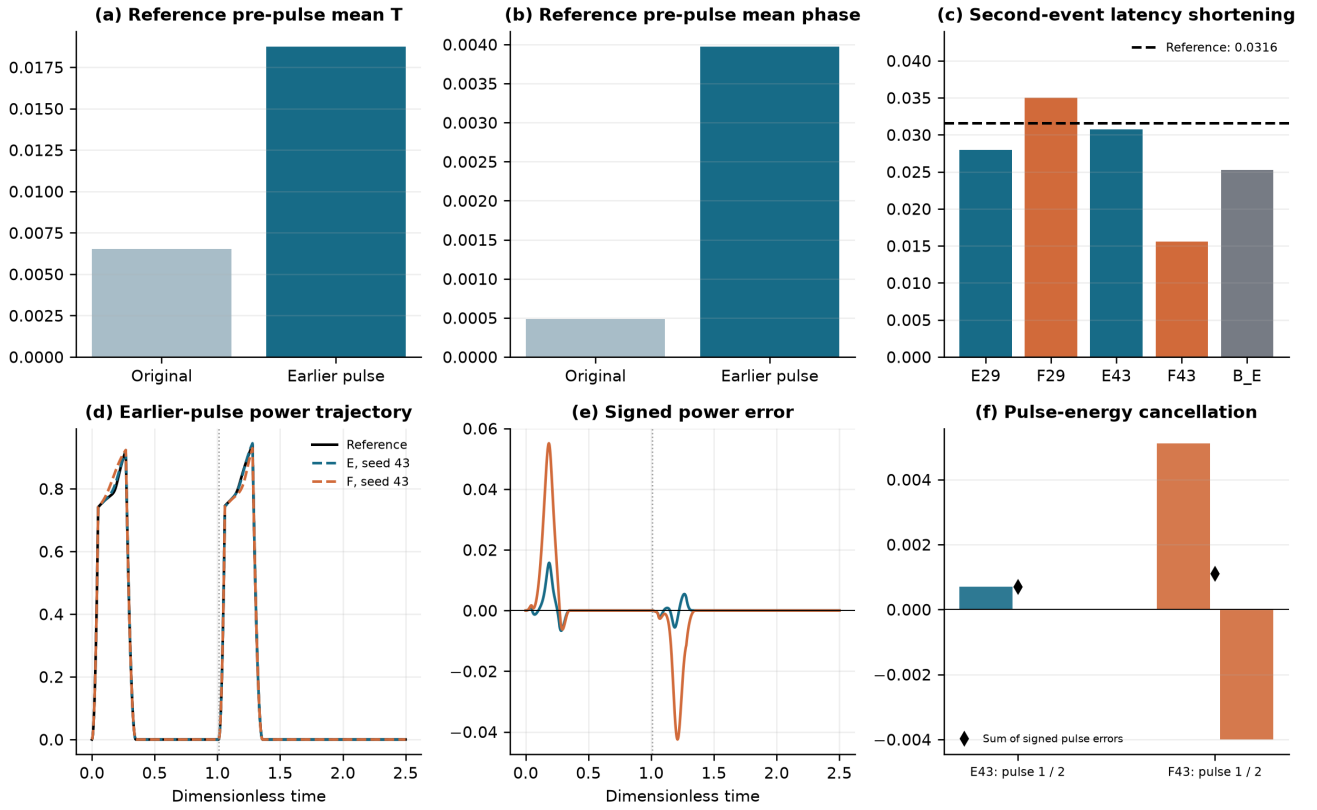


Figure 5. Panels a-b show reference ROI means before the respective second pulses; panel c uses only saved event times and is report-only. Panels d-f show the earlier-pulse case for seed 43, including signed errors and their pulse integrals. The energy diamond is the sum, not the sum of absolute errors. Sources: saved reference-history, complete-events, per-pulse tables and power traces. No state or trajectory was generated for this figure.

5.5 Historical event and energy counterexamples

Under the earlier pulse, E has first/second onset errors 0.003333/0.002775 for seed 29 and 0.003490/0.004440 for seed 43. All are below 0.005. First-cycle recalls remain 0.866607 and 0.875507, below 0.9, so neither model meets all project-defined two-cycle event requirements. F29 has a smaller second-cycle timing error, 0.0008375, than E29. E also has larger tail-phase RMS in both earlier-pulse pairs; seed 43 has a slightly larger second-cycle temperature error. Figure 6 displays these adverse outcomes together with the original-protocol event failures.

Energy integration introduces another non-equivalence. Earlier-pulse F43 has opposite signed pulse-energy errors, +0.00511674 and -0.00400099. Their cancellation gives only 0.25736% total energy error despite 3.09469% power-trajectory NRMSE. E43 also exhibits cancellation, so the phenomenon is not unique to the soft model. On the original protocol, F29 has a smaller energy error than E29 even though both current and power trajectories favor E. These observations justify retaining trajectory and signed per-pulse measures rather than interpreting integrated energy as a sufficient device score.

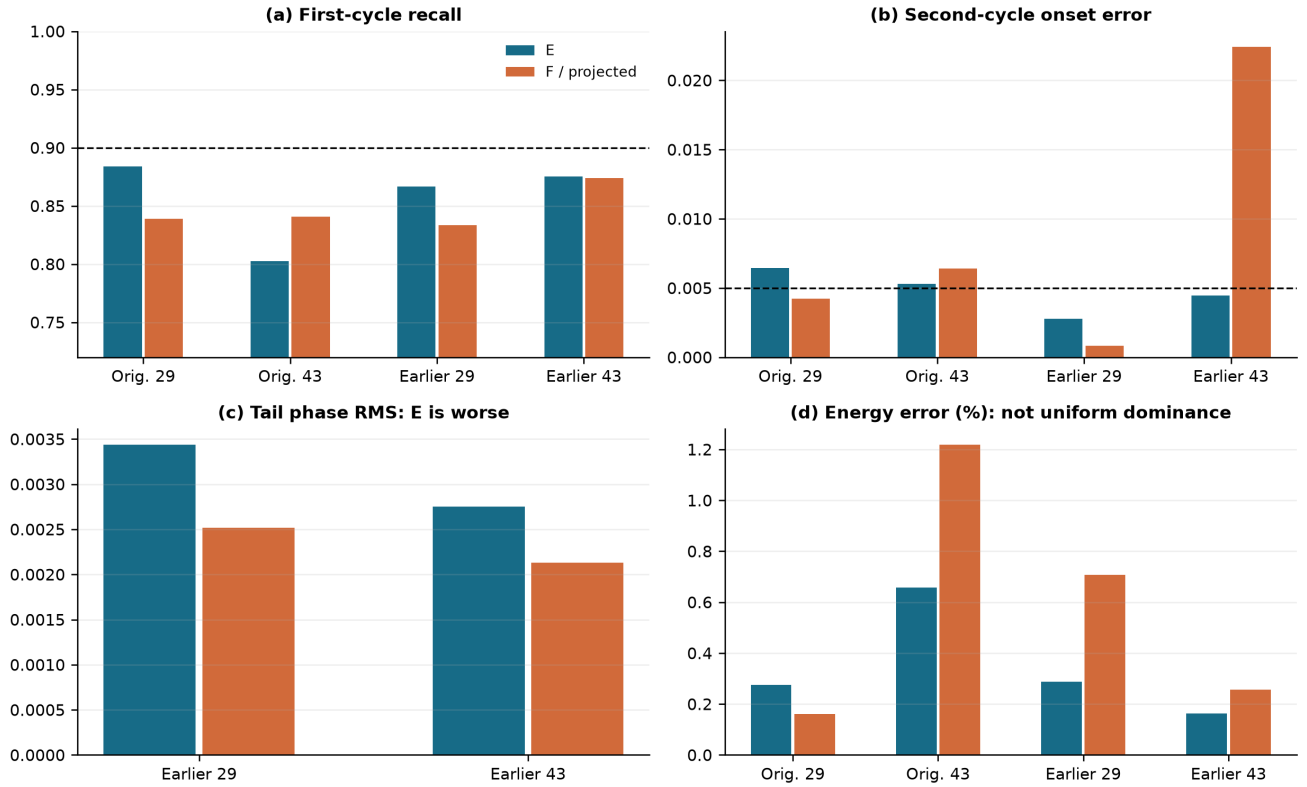


Figure 6. Dashed lines indicate the inherited recall and timing criteria, not revised thresholds. Tail errors refer to the earlier-pulse case after the second recovery cycle. The integrated-energy ranking reverses in the original seed-29 pair. All eight learned endpoints fail at least one strict event requirement. Complete precision, mass, peak, locality and recovery values accompany the event table in Supplement S5.

5.6 Phase representation: continuation, capacity and the frozen interface gate

The six prescribed coupled PINN continuations and their own electrical projections were completed. Table 3 and Figure 7 retain the parent and continued control alongside both residual heads, which add the same number of trainable parameters. E_I additionally stores a frozen copy of the parent phase head. No new reference enters optimization or checkpoint selection.

Table 3. New-head decisions under the original and refined references. A strict capability gain is separate from the phase/device criteria. Gate independence additionally compares E_I with E_R , with equal trainable counts and the extra frozen-gate cost disclosed.

Reference	Seed	Role	Matched phase	Matched device	Strict increment	Gate independent
old	29	E_R	no	no	no	not a gate
old	29	E_I	no	no	no	no
old	43	E_R	no	no	no	not a gate
old	43	E_I	no	no	no	no
refined	29	E_R	no	no	no	not a gate
refined	29	E_I	no	no	no	no
refined	43	E_R	no	no	no	not a gate
refined	43	E_I	no	no	yes	yes

For seed 29, unchanged continuation changes raw phase RMS from 0.01900241 to 0.01754883. This control is necessary because the new heads receive additional optimization.

E_R changes phase RMS by +0.97% and power NRMSE by -3.59% relative to E_C (negative is improvement). Its cycle recalls are 0.886097/0.965721, with onset errors 0.005417/0.000275.

E_I changes phase RMS by -1.58% and power NRMSE by -7.54% relative to E_C (negative is improvement). Its cycle recalls are 0.864128/0.963498, with onset errors 0.001575/0.001525.

For seed 43, unchanged continuation changes raw phase RMS from 0.01861922 to 0.01696354. This control is necessary because the new heads receive additional optimization.

E_R changes phase RMS by -0.61% and power NRMSE by -4.43% relative to E_C (negative is improvement). Its cycle recalls are 0.892181/0.959395, with onset errors 0.003300/0.000783.

E_I changes phase RMS by +2.04% and power NRMSE by -7.24% relative to E_C (negative is improvement). Its cycle recalls are 0.898603/0.962301, with onset errors 0.003750/0.002287.

Neither added head satisfies a predefined phase, device or strict increment against E_C in either parent under the original reference. The additional equal-parameter gate criterion is met in neither parent under the original reference and in 43 under the refined reference. A result in only one parent or under only one reference is a bounded development signal. It is not a clean initialization confirmation or a reason to pool different categories across seeds. No independent gate contribution is established under the original reference. The gate is retained as a tested counterfactual rather than added to the main contribution list. The absence of an original-reference increment limits the proposed representation-efficiency explanation without proving that phase representation never matters. The reference-specific strict result is examined separately below.

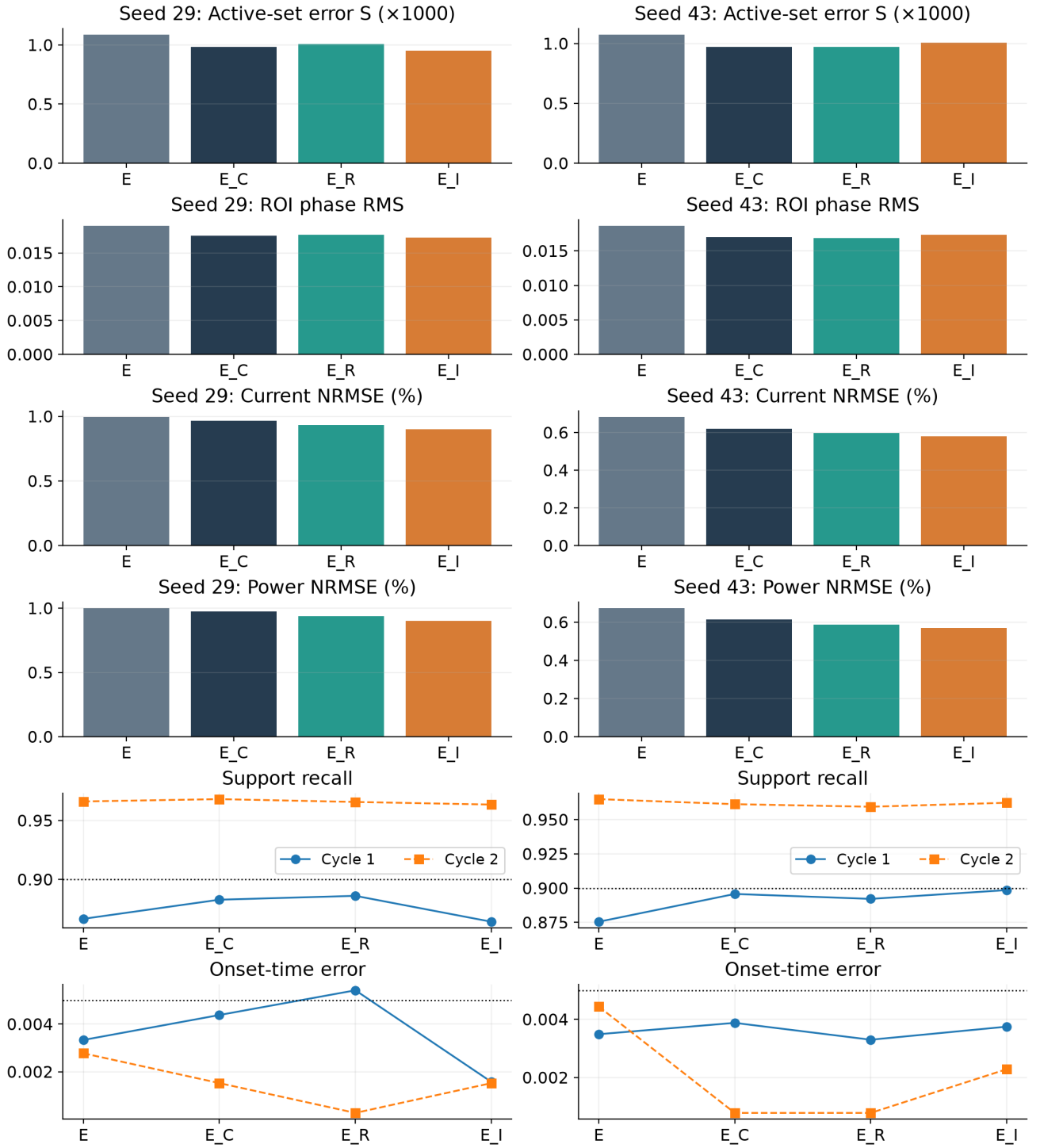


Figure 7. Both parents are displayed separately. All metrics use the original reference; the two curves in the bottom rows show the two cycles. Dashed horizontal lines retain the original criteria. Update and complete-evaluation caps are matched. E_C retains the original trainable count; E_R/E_I each add 1217 trainable parameters, and E_I additionally evaluates and differentiates a frozen gate. Supplement S10 lists actual counts and both-reference tables.

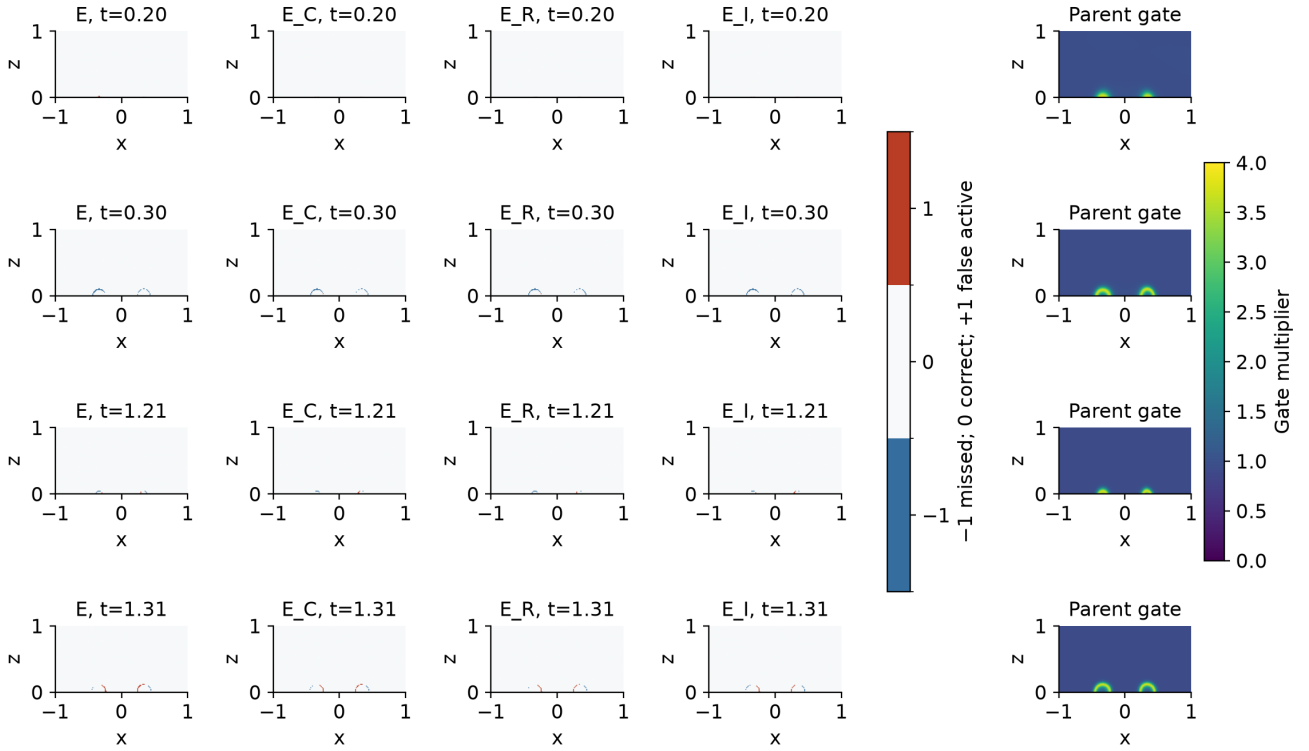


Figure 8. The four display times were fixed before training (0.20, 0.30, 1.21, 1.31); the complete domain is shown. Blue marks missed activity, rust marks false activity and white marks agreement. The final column is the frozen normalized gate. It is not selected from reference errors. The same complete plot for seed 43 appears in Supplement S10. These maps explain localization descriptively; they do not create an additional training or selection criterion.

5.7 Fixed-model reference sensitivity and array-only reproduction

With fixed historical predictions, the device criterion against the projected soft control passes in 4/4 pairs under the original references and 4/4 under the refined references. The changed historical A/B decisions are: none in the four E/soft and E/interpolant comparisons. A changed gate limits that specific threshold claim; a retained direction or gate supports only the tested temporal perturbation. Full errors, each reference denominator and fixed-old-denominator diagnostics are retained in the scoring records. The original reference discrepancy is $\delta_S = 1.6875e-05$ and ROI phase RMS $\delta = 0.00039034515$. The shorter reference discrepancy is $\delta_S = 1.671875e-05$ and ROI phase RMS $\delta = 0.00032183192$. The historically narrow seed-43 S margin changes from $4.4453125e-06$ to $7.9921875e-06$. The sufficient triangle-bound condition alone cannot certify this narrow margin; its retention follows from actual rescoring of these two references, not from a continuum argument.

Strict two-cycle success holds for 0 of the sixteen objects under the original references and 1 under the refined references. Original-reference successes: none. Refined-reference successes: shorter/43/E_I. Historical and newly developed objects are not pooled to estimate a success probability, and a reference-induced status change is not new model capability.

Table 4. Reference discrepancies in the exact scoring measures. These are changes between numerical references, not model errors against a continuum truth.

Protocol	delta S	delta phase	delta T/0.45	delta V	delta I RMS	delta power RMS	delta q RMS
original	1.6875e-05	0.000390345	0.000447933	7.21658e-05	0.000265731	0.000187004	0.000705345
shorter	1.67187e-05	0.000321832	0.00039944	5.49345e-05	0.000204478	0.000144601	0.000529945

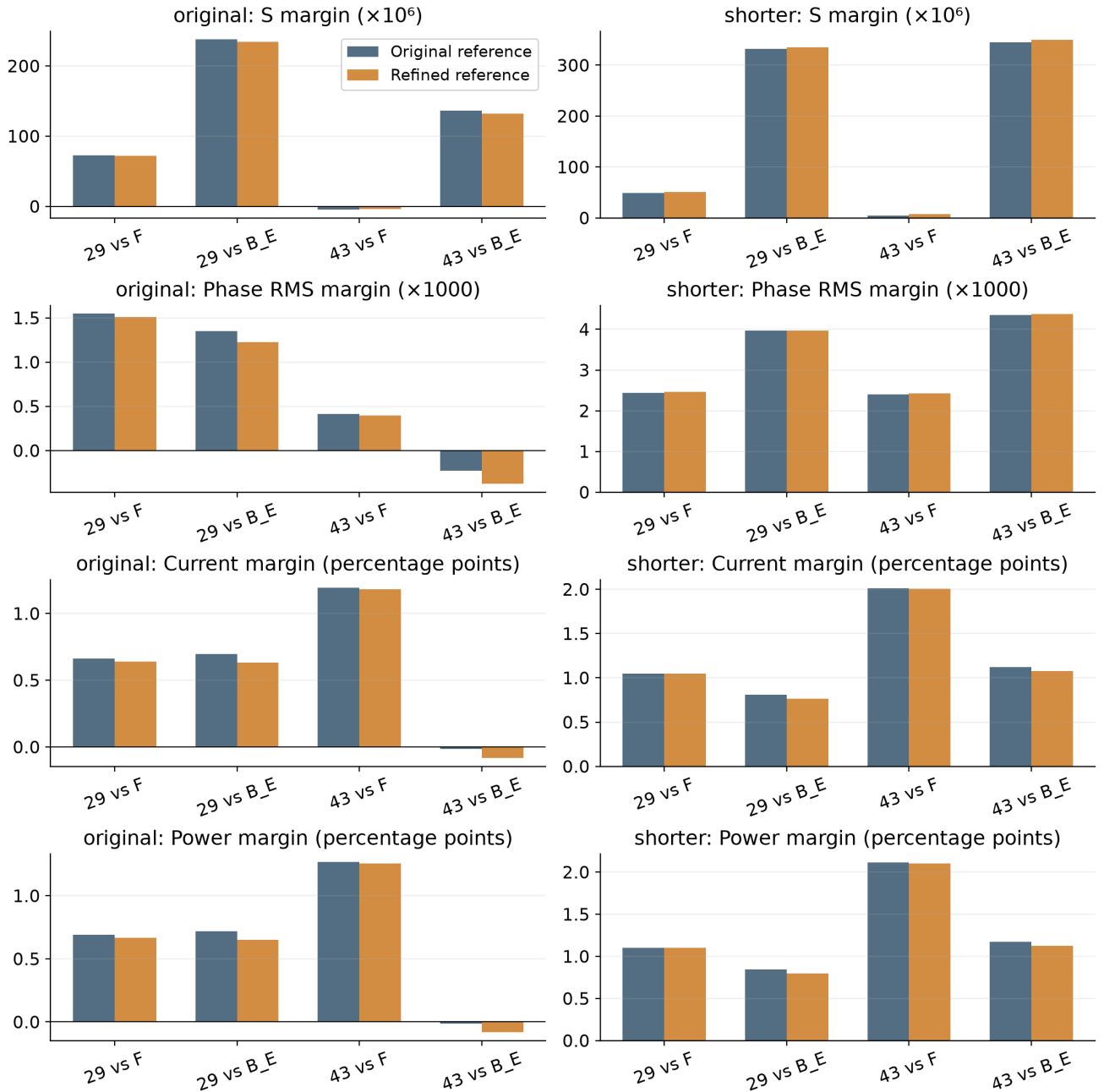


Figure 9. Original and refined relative-gain margins for the same historical E/F and E/B_E arrays, including both primary device quantities. Positive margins satisfy the relative component only; original absolute tolerances and noninferiority still enter the full decision. Phase and symmetric-difference bounds use the same fixed measure. Device bars use each reference denominator; the fixed-original-denominator version for the triangle bound is supplied separately. No new network is selected from these plots.

For shorter/43/E_I, cycle 1 recall moves from 0.898602974 to 0.902063024, although predicted active mass remains 0.000659453125. Reference active mass changes from 0.0006934375 to 0.00068921875, and overlap from 0.000623125 to 0.00062171875. The

same cycle's onset error changes from 0.003750000 to 0.004583333. This is a reference-specific crossing of the predefined strict criterion; it does not establish a gate contribution that survives both reference choices.

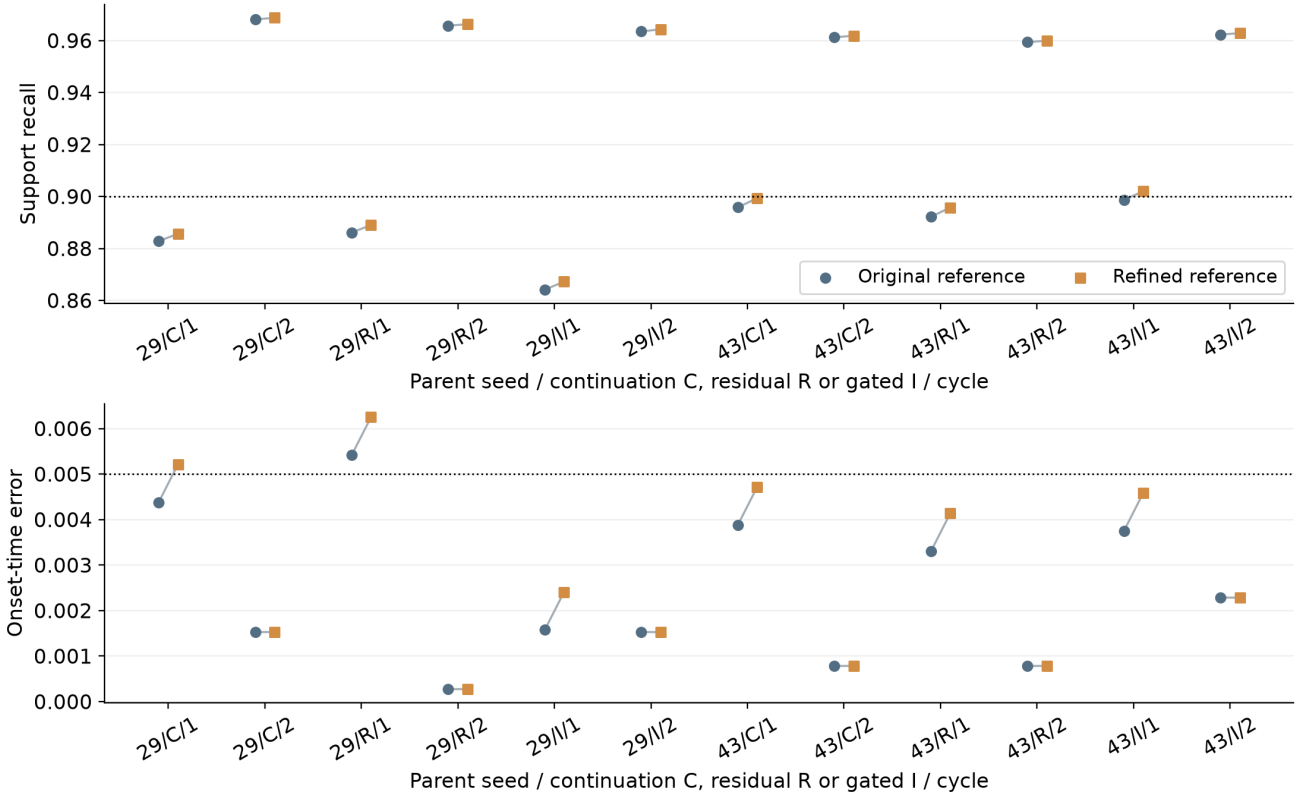


Figure 10. Both cycles of all six fixed new endpoints are displayed. Paired points change the reference only, preserving the predicted fields, event times and active masses. Dotted lines are the original recall and timing requirements; passing these two quantities alone does not replace the complete strict rule. The trace for seed 43, gated head, cycle 1 shows the reference-sensitive crossing without suppressing the other endpoints.

The clean-directory NumPy evaluation reproduces all ten old objects' saved metrics, complete event summaries and A/B decisions, and then evaluates all six new objects against both references. This establishes array-level reproducibility of the submitted results. It neither retrains the models nor proves independence of the original solver implementation. The local archive contains the full scoring inputs; public release is still an author-controlled step.

6. Discussion

The most direct interpretation is that how the electrical constraint participates in training affects the learned evolving state, even when electrical repair is available to every method at inference. The repair controls show that much of an unprojected soft model's apparent device error can disappear without changing phase. The residual advantage after common projection, its persistence against a full-spatial penalty in development, and the clean paired results together support the specified training package. They do not establish that an isolated implicit gradient, rather than differences in parameterization, enforcement times or optimization, is the unique cause.

This evidence is stronger than comparison with an uncorrected weak baseline, but narrower than a general claim that exact elimination always dominates soft penalties. Only a finite set of penalty recipes and a fixed optimization budget were tested. D_E also limits the role assigned to the remaining interior residuals. A coupled PINN can have an

evidenced algorithmic advantage without every retained term having a demonstrated independent increment; the latter claim is explicitly withheld here.

The application scope is equally specific. Two-dimensional feedback, localized electrodes and pulse-dependent residual states make this a device-inspired computational test rather than an abstract one-variable example. They do not make it a calibrated oxide or chalcogenide memory model. A material-facing claim would require an internally consistent material model, parameter provenance and independent physical validation. Those are absent and are not supplied by citing a more detailed device study.

The measurements also have numerical limits. All formal comparisons use the fixed reference discretization and previously fixed selection rules. Shared conservation and dissipation identities express consistency of the numerical operator, not independent accuracy evidence. The narrow S margin and threshold-sensitive events especially should not be promoted to continuum claims. The new fixed-prediction comparison quantifies one temporal-reference perturbation and its actual decision changes in Section 5.7. It remains a fixed-spatial-discretization study; unchanged rankings cannot be promoted to continuum accuracy. The new phase-head controls address one representation hypothesis without establishing the independent necessity of the retained PDE terms.

The added phase heads yield small and mixed changes beyond continuation. This constrains the tested capacity explanation while preserving the reference-specific strict crossing as a numerical sensitivity result. It does not justify escalating the gate into the core method or declaring phase capacity irrelevant.

The explicit material mapping in Supplement S12 separates state, geometry, transport, latent heat and kinetics assumptions from validation. It also retains a theoretical oxide counterpoint rather than treating Joule heating as the sole possible transition mechanism [14].

7. Conclusions

A partially eliminated hybrid PINN couples neural temperature and phase fields to an implicit quasi-static electrical layer and a consistent local Joule source. Under two complete pulse protocols and two clean initialization pairs, training-time elimination improves current and power reconstruction beyond a common post-training electrical solve. The comparison includes a same-solver interpolant and development counterfactuals that limit explanations based solely on readout repair or electrical spatial sampling. Additional matched phase-head continuations and a fixed-model time-reference perturbation delimit capacity, gate and numerical-sensitivity claims rather than silently expanding the method. Continuous residual states also distinguish threshold recovery from complete relaxation. The supported benefit is conditional on the stated numerical task and training recipe: the original eight learned states do not meet the complete strict two-cycle requirements. Strict two-cycle success holds for 0 of the sixteen objects under the original references and 1 under the refined references. Original-reference successes: none. Refined-reference successes: shorter/43/E_I. Historical and newly developed objects are not pooled to estimate a success probability, and a reference-induced status change is not new model capability. Independent necessity of the remaining PDE terms, isolated implicit-gradient causality and material-calibrated performance remain unestablished.

Data, code and declarations

The versioned research evidence is available in the [PINN-PCM-SCI repository](#), including source, configurations, checkpoints, curated result tables and selected traces. The preceding manuscript-only version is separately identified by [commit 4081ba09](#). This is the local experimental revision dated 16 September 2026; it does not reuse either published version identifier as its own. The present revision adds six phase-head PINN continuations, their projected outputs, two time-refined numerical references and array-only rescoring. These new results and their complete local scoring archive have not been publicly uploaded. The earlier curated public package remains distinct from this complete local revision. Supplement S7 identifies the files needed to rebuild these manuscript figures without executing a scientific model and the separate scope of a full numerical reproduction. The local submission archive now contains the four references, sixteen fixed prediction/readout objects, two sparse observation bundles, accepted neural/optimizer states and portable evaluator. A clean-directory isolated-Python rescore is recorded separately; public release and a persistent dataset identifier remain submission preparation items.

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